



Growing Trees — English Worksheet

Topic: Two counters in one loop · **Course:** 3D Coordinates · **Level:** Extension · **Time:** about 40 minutes

TYPE **DEL** BEFORE EVERY **RUN**. EVERY TIME.

Every **???** is one short thing for you to work out. There are no answers on this sheet.

1 · One tree, two fills

A tree is a thin pole and a fat box.

```
blocks.fill(OAK_LOG,
    pos(x, 0, z),
    pos(x, trunk_h, z), REPLACE)
```

In the trunk fill, which of x, y, z is different between the two corners?

So what shape do you get when only one of the three changes?


```
blocks.fill(OAK_LEAVES,
    pos(x - leaf, trunk_h + 1, z - leaf),
    pos(x + leaf, trunk_h + 3, z + leaf), REPLACE)
```

One corner uses minus. The other uses plus. Why?

If leaf is 2, how wide is the leaf box? Count the middle block too.

2 · Two counters

```
trunk_h = trunk_h + 2
leaf = leaf + 1
x = x + 8
```

Start values: `x = 4` , `trunk_h = 4` , `leaf = 2` .

Fill the table.

I	X	TRUNK_H	LEAF	LEAF WIDTH
1	4	4	2	5
2				
3				
4				

Look down the columns.

`trunk_h` goes up by 2. `leaf` goes up by 1. Which tree is tallest, and which has the widest leaves?

3 · Build it 🌳

TEMPLATE

```
def on_chat():
    x = 4
    z = 8
    trunk_h = 4
    leaf = 2
    for i in range(1, ???):
        blocks.fill(OAK_LOG,
                    pos(x, 0, z), pos(x, ???, z), REPLACE)
        blocks.fill(OAK_LEAVES,
                    pos(x - ???, trunk_h + 1, z - ???),
                    pos(x + ???, trunk_h + 3, z + ???), REPLACE)
        trunk_h = trunk_h + ???
        leaf = leaf + ???
        x = x + ???
player.on_chat("trees", on_chat)
```

CAREFUL

The three counter lines go at the **end** of the loop, after both fills. Put them first and your first tree comes out wrong.

Before you run: how many trees will you get?

WORKS WHEN

three trees stand in a row, each taller and wider than the one before.

ALL THE TREES ARE THE SAME?

Your counter lines are outside the loop. Check the spaces at the front.

ONLY ONE TREE?

`x` is not changing, so every tree is built in the same place.

4 · The gap

Tree 3 has **leaf = 4**, so its leaves are 9 wide.

Trees are 8 apart. Is 8 enough space for a 9 wide tree? Try it and look.

Change the 8 to a smaller number. Run it. What happens to the leaves?

YOU UNDERSTAND IT WHEN

you can say the smallest gap that keeps every tree separate.

5 · Clean up

Your trees reach $x = 24$ and $y = 11$.

TEMPLATE

```
blocks.fill(AIR,  
    pos(0, 0, 0), pos(???, ???, ???), REPLACE)
```

Your old `del` used 15. Why is that not big enough now?

6 · Show Your Work 📷 🎥

Record **one video**, one take, no stopping. A phone is fine. Show these in order:

1. Type **del** , then **trees** . Walk past all three trees.
2. Show the trunk fill and say why it makes a pole.
3. Show the leaf fill and point at the minus and the plus.
4. Show your three counter lines and say what each one grows.
5. Show your filled table and read row 3 out loud.

Try to use these words: **loop, counter, trunk, leaves, gap.**

Plan what you will say before you record.

Submit ✓

Send this worksheet and your video to teacher on KakaoTalk.